

Rajesh Kodaganti

Software Development Engineer

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SUMMARY

Software Development Engineer with a Master of Science in Computer Engineering and hands-on experience designing and building scalable, distributed, cloud-native systems in Python. Skilled in object-oriented design, microservices, REST API orchestration, CI/CD, production monitoring and troubleshooting, RAG, document processing, and secure infrastructure as code. Delivered a six-phase outage-management pipeline supporting Severity 0-2 incidents across 70+ Microsoft Identity services.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, R, C, Embedded C, HTML

Software Engineering: FastAPI, REST API Orchestration, Microservices, Object-Oriented Design, Event-Driven Architecture, ETL & Data Pipelines, Git/GitHub, Testing & Troubleshooting

Azure & DevOps: Azure Functions, Logic Apps, Service Bus, Blob Storage, Key Vault, Application Insights, Entra ID, Bicep, CI/CD, Docker, Apache Airflow

AI & Machine Learning: Azure OpenAI, Azure AI Search, Azure Document Intelligence, RAG, LLMs, NLP, Prompt Engineering, PyTorch, TensorFlow, Model Context Protocol

EXPERIENCE

Microsoft

Oct 2025 – Present

AI Software Engineer (Contract) | Redmond, Washington

- Designed and developed a scalable, six-phase outage-management pipeline automating incident ingestion, enrichment, classification, validation, and reporting for Severity 0-2 incidents across 70+ Microsoft Identity services
- Built Python Azure Functions and KQL workflows for IcM data processing and integrated Azure OpenAI to classify root causes and recurring outage patterns
- Automated DRI notifications, escalation workflows, and Excel and Word reporting through Logic Apps
- Provisioned secure Azure infrastructure using Bicep, managed identity, Key Vault, Blob Storage, Service Bus, Application Insights, and Entra ID; delivered a monitoring dashboard, runbooks, troubleshooting guidance, and knowledge-transfer documentation

Quadrant Technologies

Aug 2025 – Present

AI Software Engineer | Redmond, Washington

- Architected and deployed scalable AI solutions with Azure OpenAI and Azure Cognitive Services for threat intelligence and automated mitigation workflows
- Implemented Azure Document Intelligence workflows to extract and structure unstructured data
- Developed RAG architectures integrating Azure AI Search with LLMs for accurate, low-latency retrieval across security vulnerability reports
- Built Sentinel, an autonomous DevOps orchestrator bridging Jira and GitHub with Azure OpenAI, GitHub Copilot, and MCP to take tickets from To Do to Done with minimal human intervention

Quadrant Technologies

Jul 2025 – Aug 2025

Artificial Intelligence Trainee | Redmond, Washington

- Designed ETL processes for security logs and vulnerability reports and developed endpoint and network-security data-ingestion systems to enhance risk analysis
- Applied machine learning to threat-intelligence data and created cloud-security monitoring dashboards supporting compliance and data integrity

QikCell by tickioT

Jan 2025 – May 2025

Assistant to Solutions Engineer | Los Angeles, California

- Served as the primary technical liaison for client field trials, bridging antenna and PCB hardware constraints with AI software requirements

- Customized IoT demonstrations using Python to simulate client use cases and showcase AI-enhanced signal processing, predictive maintenance, and operational efficiency

ColorOS

Jun 2018 – Jul 2022

User Tester (Freelance) | India (Remote)

- Conducted user testing for ColorOS Beta and Alpha versions (5.0 to 12.1)
- Documented defects and proposed product improvements in collaboration with cross-functional teams

Pantech ProEd Pvt Ltd

Jun 2020 – Jul 2020

Student Intern | Hyderabad, India

- Contributed to the development of a high-speed 64-bit binary comparator
- Optimized design parameters using Verilog and VHDL

PeopleLink Unified Communications Pvt. Ltd.

Jun 2018 – May 2019

IoT Engineer in R&D | Hyderabad, India

- Executed end-to-end testing of connected embedded devices and developed automated Python test scripts that reduced manual testing time by 20%
- Analyzed TCP/IP and UDP communication protocols to troubleshoot real-time latency issues; collaboration with R&D cut post-launch defects by 20%

EDUCATION

California State University, Northridge

Aug 2022 – May 2025

Master of Science - MS, Computer Engineering

JB Institute of Engineering and Technology

Jul 2019 – Jun 2022

Bachelor of Technology, Electronics and Communication Engineering | Score: 7.45

PROJECTS

IcM Outage Automation - Microsoft Identity LiveSite

Python, Azure Functions, KQL, Azure OpenAI, Logic Apps, Bicep

A six-phase outage-management system that automates incident ingestion, enrichment, classification, validation, escalation, and reporting for Microsoft Identity LiveSite operations.

- Implemented six phases covering incident ingestion, enrichment, classification, validation, escalation, and reporting.
- Integrated Azure OpenAI for root-cause and recurring-pattern classification.
- Automated DRI notifications and generated Excel and Word reports for incident review.

JobPilot: AI-Powered Job Application Co-Pilot | [link](#)

JavaScript, Python, FastAPI, Cloudflare Workers, Anthropic API, AES-GCM

An AI-powered job application co-pilot built with JavaScript, Python, and FastAPI. JobPilot aggregates listings from Adzuna, JSearch, and USAJobs and supports AI-assisted application tailoring, GitHub Pages deployment, a Cloudflare Worker proxy, and client-side AES-GCM encryption.

- Aggregates job listings from Adzuna, JSearch, and USAJobs into a single search experience.
- Static site generated with Python and hosted on GitHub Pages, paired with a thin Cloudflare Worker proxy to eliminate a traditional backend database.
- Zero-trust, client-side security model: Anthropic API keys and user resumes are encrypted with AES-GCM and stored via direct Google Drive integration for complete user privacy.

Sentinel: Autonomous DevOps Orchestrator | [link](#)

Azure OpenAI, GitHub Copilot, MCP, JavaScript, DevOps

An AI-driven bridge between Jira and GitHub designed to automate the manual overhead of the DevOps lifecycle, taking tickets from 'To Do' to 'Done' with minimal human intervention. Associated with Quadrant Technologies.

- Autonomous polling that scans Jira boards every 30 seconds for new tickets.
- AI-powered implementation using Azure OpenAI and GitHub Copilot to analyze repository structures and generate context-aware code fixes.
- Automated PR management: creates feature branches, triggers Copilot, and manages the approval/merge flow once CI/CD checks pass.

GasOps MTR extraction | [link](#)

Azure Document Intelligence, OpenAI API, Python, FastAPI, API Testing

Built a JSON extraction pipeline at Quadrant Technologies using Azure Document Intelligence and the OpenAI API for AI/OCR orchestration and structured downstream processing.

- Implemented Azure Document Intelligence OCR and AI mapping supporting Azure OpenAI
- Built certificate and token-based API fetch by HeatNumber
- Created robust JSON extraction from model responses and a backwards-compatible CLI shim

Image Denoising using Deep Learning | [link](#)

Python, TensorFlow, PyTorch

Developed ConvIR, a deep learning framework for image denoising targeting additive white Gaussian noise (AWGN) using a dual-domain learning approach. Associated with California State University, Northridge.

- Implemented a composite loss function incorporating pixel-wise MSE, FFT-based spectral loss, and noise estimation regularization for spatial and frequency fidelity.
- Designed a UNet-inspired convolutional network with skip connections, multi-shape attention, and frequency-aware branches, trained across noise intensities ($\sigma=1.5$ to $\sigma=25$).
- Achieved PSNR gains exceeding 12 dB at high noise levels and a 95% reduction in residual noise under low-noise conditions.

Pre-owned Cars pricing forecast using R language

R, Data Preparation

Developed a Random Forest Regression model in R for pre-owned car price forecasting.

- Achieved 85.67% accuracy on a dataset of 48,904 rows
- Conducted extensive data preprocessing including handling missing values and categorical features

1X3 Router Design and Verification

Verilog, RTL Design

Designed and verified a 1x3 router using Verilog HDL.

- Implemented synthesis constraints in Verilog
- Performed verification using testbenches in Modelsim and synthesis in Quartus Prime

Raspberry Pi Applications for Assistive Technologies | [link](#)

Raspberry Pi, Python, Tesseract OCR

Created reading aids for visually impaired individuals using Raspberry Pi and OCR technology.

- Utilized Tesseract library and Python to convert document images into audio format using both online and offline OCR methods
- Integrated camera modules for real-time text capture in educational settings
- Published "Readers for the Visually Impaired Using Raspberry Pi" in IJRAR, May 2022

PeopleLink "Look At Me" Controller

PCB Design, Embedded C

Spearheaded circuit design for a smart camera controller for conference rooms.

- Developed wired and wireless models with auto-focus capabilities on active speakers
- Simplified installation and operation for collaborative environments

CERTIFICATIONS

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|---|---|
| • Career Essentials in Sustainable Tech — Microsoft, Sep 2024 | • Azure Data Scientist Associate — Microsoft, Feb 2026 |
| • Microsoft Azure AI Essentials Professional Certificate — Microsoft, Jul 2025 | • Claude Code in Action — Anthropic, Feb 2026 |
| • Career Essentials in GitHub Copilot Professional Certificate — GitHub, Jul 2025 | • Introduction to Model Context Protocol — Anthropic, Feb 2026 |
| • Docker Foundations Professional Certificate — Docker, Jul 2025 | • Model Context Protocol: Advanced Topics — Anthropic, Feb 2026 |
| • Apache Airflow 3 Fundamentals — Astronomer, Oct 2025 | • Claude in Amazon Bedrock — Anthropic, Feb 2026 |
| • DAG Authoring for Apache Airflow 3 — Astronomer, Oct 2025 | • Building with the Claude API — Anthropic, Feb 2026 |
| • Fabric Data Engineer Associate — Microsoft, Nov 2025 | • Claude with Google Cloud's Vertex AI — Anthropic, Feb 2026 |
| • Azure AI Engineer Associate — Microsoft, Dec 2025 | • Career Essentials in Generative AI — Microsoft, Mar 2026 |
| • Microsoft Copilot for Productivity — Microsoft, Dec 2025 | • SQL AI Developer Associate — Microsoft, May 2026 |
| • Atlassian IT Service Management (ITSM) Professional Certificate — Atlassian, Dec 2025 | • AI Fluency: Framework & Foundations — Anthropic, Jun 2026 |
| • DevOps Engineer Expert — Microsoft, Jan 2026 | • Introduction to Claude Cowork — Anthropic, Jun 2026 |
| • Azure Administrator Associate — Microsoft, Jan 2026 | • AI Capabilities and Limitations — Anthropic, Jun 2026 |